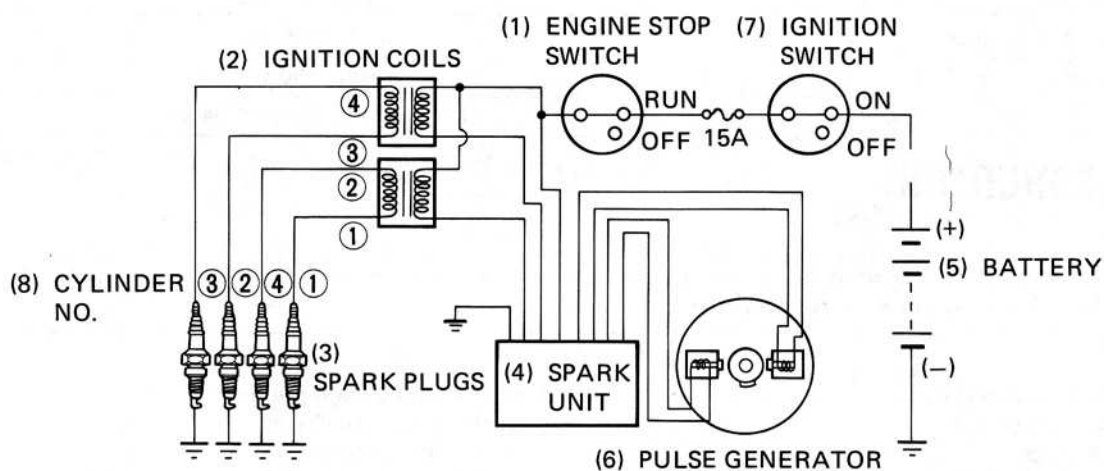
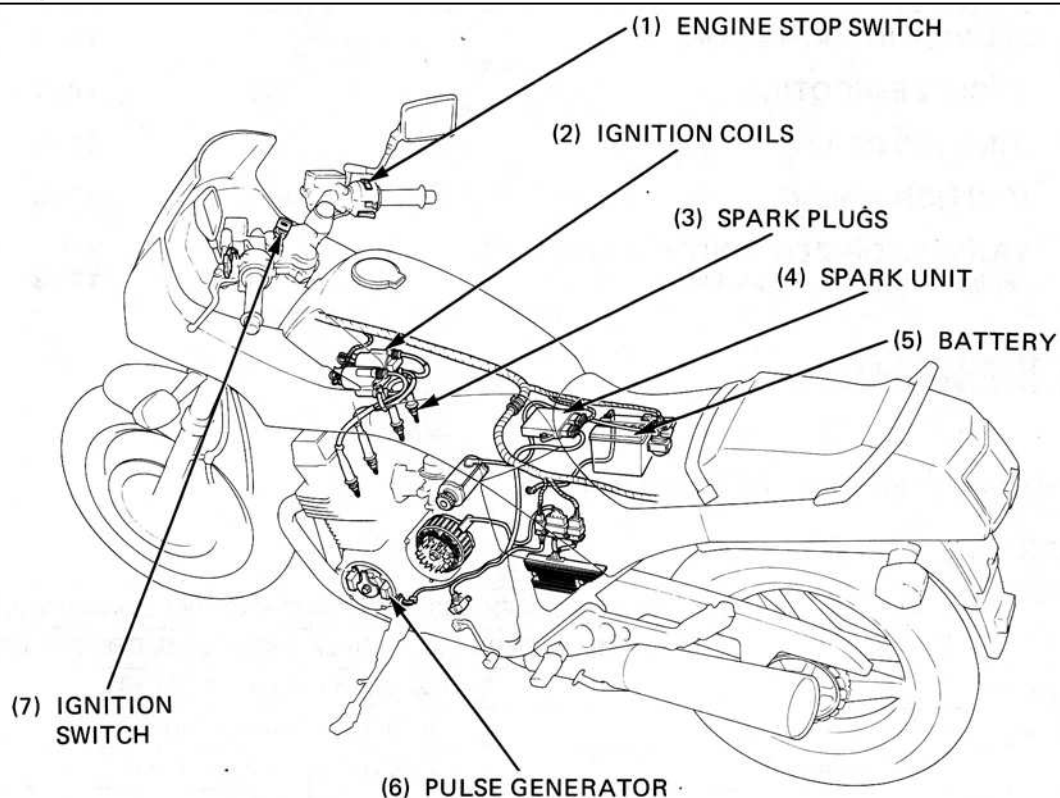




17. Ignition System





SERVICE INFORMATION	17-1	IGNITION TIMING	17-3
TROUBLESHOOTING	17-1	TRANSISTORIZED IGNITION SYSTEM	17-3
IGNITION COIL	17-2	(Pulse Generator, Spark Unit)	

SERVICE INFORMATION

GENERAL

A transistorized ignition system is used and it cannot be adjusted.

SPECIFICATIONS

< > : U

Spark plug	ND	X22EPR-U9, X24EPR-U9, X27EPR-U9 <X22EP-U9, X24EP-U9, X27EP-U9>
	NGK	DPR7EA-9, DPR8EA-9, DPR9EA-9 <DP7EA-9, DP13EA-9, DP9EA-9 >
Spark plug gap		0.8-0.9 mm (0.031-0.035 in)
Ignition timing	At idle	10 BTDC/1,000min ⁻¹ (rpm)
	Full advance	32 BTDC/3,150 min ⁻¹ (rpm)
Ignition coil resistance	Primary	2.4-3.0Ω
	Secondary	13.6-15.5 kΩ
Spark plug cap resistance		3.75-6.25 kΩ
Pulse generator	Resistance	300-360Ω
	Air gap	0.5-0.9 mm (0.02-0.04 in)

TROUBLESHOOTING

NOTE:

The ignition system has two sub-systems; one for the No. 1 and No. 4 cylinders and one for No. 2 and No. 3 cylinders.

Determine which sub-system is faulty, then refer to the charts below.

Engine cranks but will not start	Engine starts but runs poorly
- Engine stop switch OFF	- Ignition primary circuit
- No spark at plugs	• Faulty ignition coil
- Faulty transistorized spark unit	• Loose or bare wire
- Faulty pulse generator	• Intermittent short circuit
	- Secondary circuit
No spark at plug	• Faulty plug
- Engine stop switch OFF	• Faulty spark plug wire
- Poorly connected, broken or shorted wires	
• Between ignition switch and engine stop switch	Timing advance incorrect
• Between spark unit and engine stop switch	- Centrifugal advancer faulty
• Between spark unit and ignition coil	- Faulty spark unit
• Between ignition coil and plug	
• Between spark unit and pulse generator	
- Faulty ignition coil	
- Faulty ignition switch	
- Faulty spark unit	
- Faulty pulse generator	



IGNITION COIL

CONTINUITY TEST

Remove the fuel tank and disconnect the ignition coil wire leads. Measure the primary coil resistance between the terminals.

RESISTANCE: 2.4-3.0 Ω

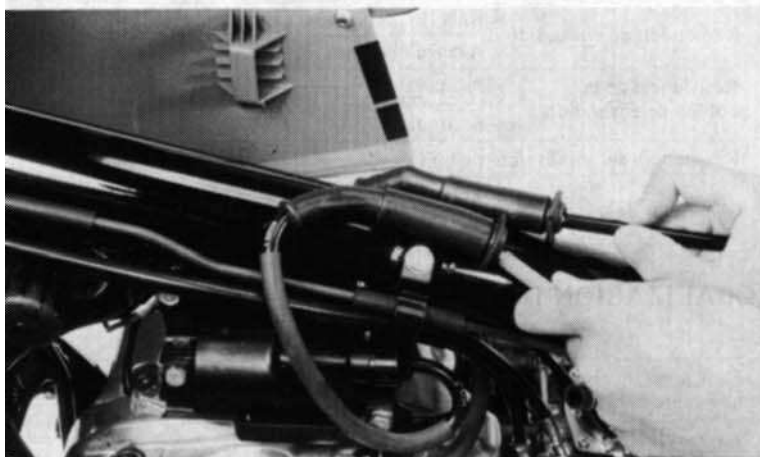
If the reading does not meet the specification, check for continuity between the primary terminal and ground. Replace the coil if there is continuity between them.



Measure the secondary coil resistance with the spark plug caps in place.

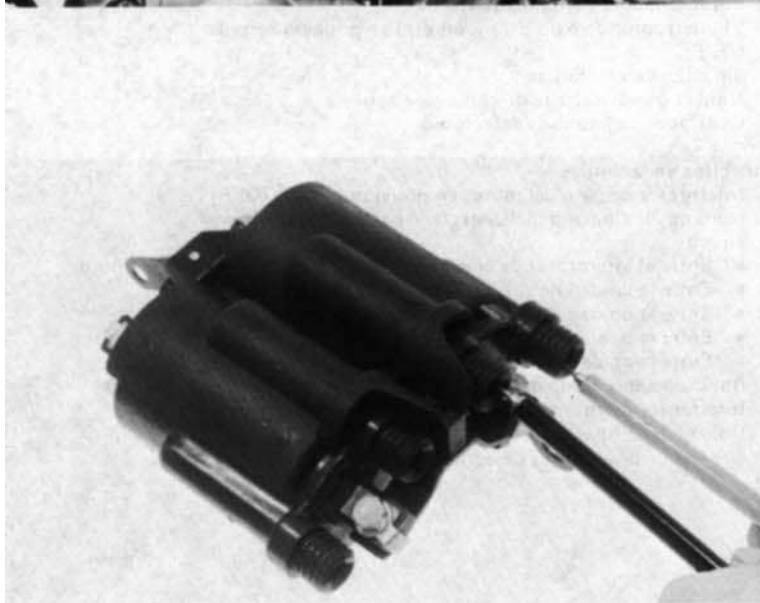
RESISTANCE: 21-28 k Ω

If the reading does not fall within the limit, remove the ignition coils by removing the bolts.



Remove the spark plug wires by loosening the wire retaining nuts. Measure the secondary coil resistance.

RESISTANCE: 13.6-15.5 k Ω





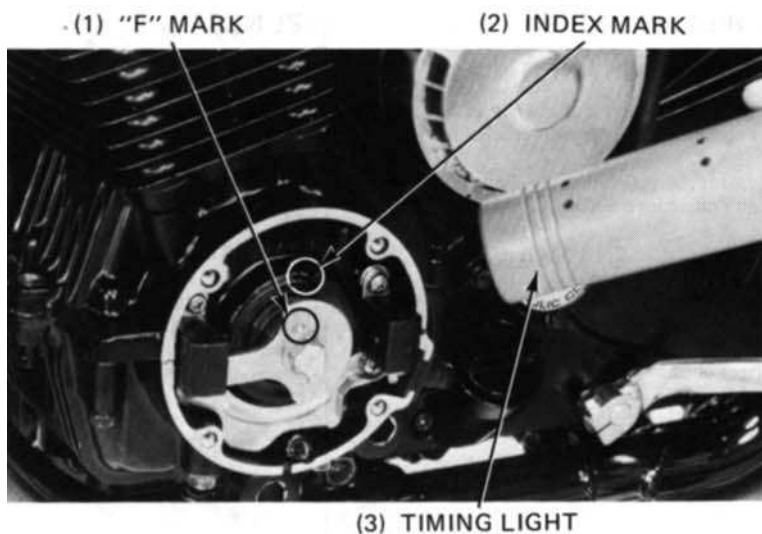
IGNITION TIMING

NOTE:

The ignition system is transistorized and cannot be adjusted. If the ignition timing is incorrect, check the spark unit and pulse generator and replace any faulty parts.

Warm up the engine.

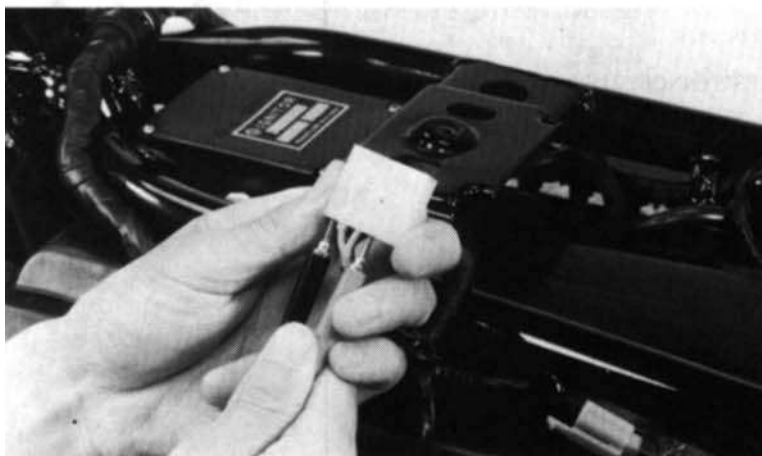
Remove the pulse generator cover.
Connect the timing light to the No. 1 or No. 4 cylinder's spark plug wire.
The timing is correct if the "F" mark aligns with the index mark on the crankcase at $1,000 \pm 100$ rpm.



TRANSISTORIZED IGNITION SYSTEM

PULSE GENERATOR TEST

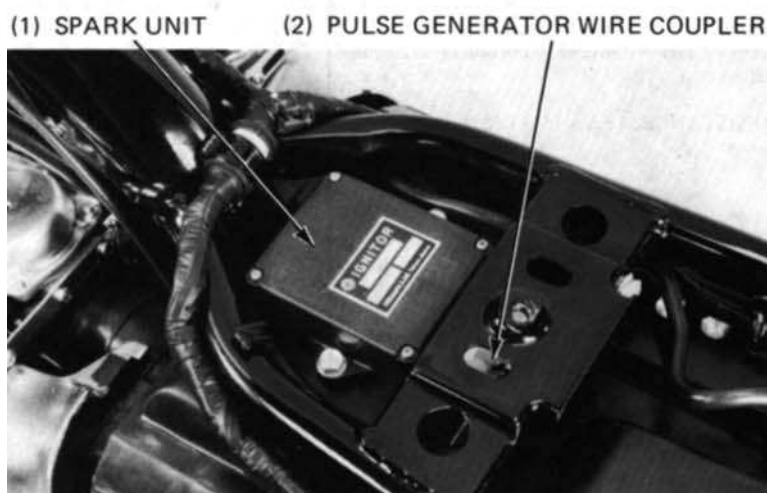
Remove the seat and fuel tank. Disconnect the pulse generator coupler and measure the coil resistance between the white and yellow wires (2 and 3 cylinders), and between the white, and blue wires (1 and 4 cylinders).



RESISTANCE: 300-360Ω

PULSE GENERATOR REPLACEMENT

Remove the seat and fuel tank.
Disconnect the pulse generator wire coupler from the spark unit.





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17. Ignition System

Remove the pulse generator cover and drive sprocket cover.

(1) PULSE GENERATOR COVER



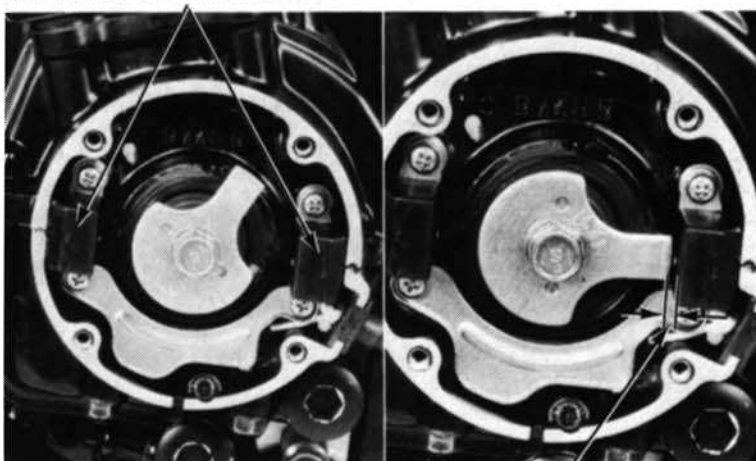
Remove the pulse generator mounting screws, pulse generators and wire guide. Install new pulse generators and wire guide.

Align the pulse rotor tip with the pulse generator magnet and measure the air gap with a feeler gauge.

AIR GAP: 0.5-0.9 mm (0.02-0.04 in)

Check the ignition timing.
Install the drive sprocket cover and pulse generator cover.

(1) PULSE GENERATORS



(2) AIR GAP

SPARK UNIT

If the pulse generators, ignition coils and wiring are good, and the ignition timing is not within specification; replace the spark units with new ones and recheck the ignition timing.

(1) SPARK UNIT

